

EDEXCEL INTERNATIONAL A LEVEL

WMA12/01 May/June 2025 Worked Solutions

May/June 2025 paper rebuilt with the worked solution after each question.

10

paper questions

WMA12

Pure 2

**Single-
paper
solutions**standalone IAL
route**Dr Eslam Ahmed**Prepared for Dr Eslam Ahmed - eliteigcse.com**P2**

PAST PAPER

WMA12/01 May/June 2025

May/June 2025 | 10 questions | 75 marks

10

questions

75

marks

Answers

worked solution
after each
question

Question 1

Binomial Expansion

- 1: (a) Find the first 4 terms, in ascending powers of x , of the binomial expansion of

$$(1 - 4x)^7$$

giving each term in simplest form.

(4)

In the series expansion of

$$(5 + kx)(1 - 4x)^7 \quad \text{where } k \text{ is a constant}$$

the coefficient of the term in x^2 is 1316

- (b) Use the answer to part (a) to find the value of k .

(2)

Worked Solution - Question 1

1. Write the binomial terms

$$(1 - 4x)^7 = 1 + 7(-4x) + \binom{7}{2}(-4x)^2 + \binom{7}{3}(-4x)^3 + \dots$$

2. Simplify

The first four terms are $1 - 28x + 336x^2 - 2240x^3$.

3. Find the x squared terms

In $(5 + kx)(1 - 4x)^7$, the x^2 coefficient comes from $5(336x^2)$ and $(kx)(-28x)$.

4. Use the given coefficient

$$5(336) - 28k = 1316, \text{ so } 1680 - 28k = 1316.$$

5. Find k

$$28k = 364, \text{ hence } k = 13.$$

Final answer

$$(a) 1 - 28x + 336x^2 - 2240x^3.$$

$$(b) k = 13.$$

Question 2

Circles

2: The line joining the points $(-2, 5)$ and $(4, 15)$ is the diameter of a circle C .

(a) Find an equation for C .

(5)

(b) Hence find the exact coordinates of the point on C that is nearest the x -axis.

(2)

Worked Solution - Question 2

Topic group

1. Find the centre

The centre is the midpoint of $(-2, 5)$ and $(4, 15)$, so it is $\left(\frac{-2+4}{2}, \frac{5+15}{2}\right) = (1, 10)$.

2. Find the radius squared

Using $(-2, 5)$, $r^2 = (-2 - 1)^2 + (5 - 10)^2 = 9 + 25 = 34$.

3. Write the circle equation

Therefore C has equation $(x - 1)^2 + (y - 10)^2 = 34$.

4. Find the nearest point to the x axis

The nearest point to the x -axis is vertically below the centre by one radius.

5. Write the coordinates

The radius is $\sqrt{34}$, so the point is $(1, 10 - \sqrt{34})$.

Final answer

(a) $(x - 1)^2 + (y - 10)^2 = 34$.

(b) $(1, 10 - \sqrt{34})$.

Question 3

Integration

3:

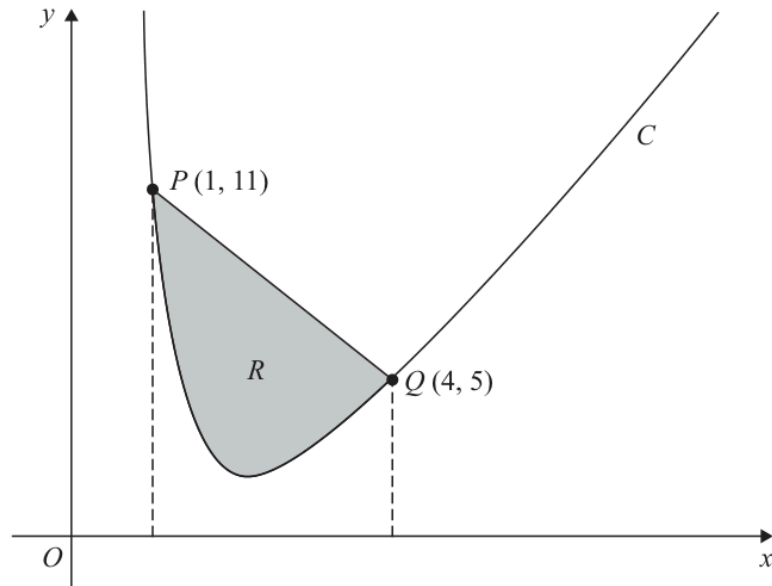


Figure 1

In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.

Figure 1 shows a sketch of the curve C with equation

$$y = 3x + \frac{16}{x^2} - 8 \quad x > 0$$

The points $P(1, 11)$ and $Q(4, 5)$ lie on C and are shown in Figure 1.

The region R , shown shaded in Figure 1, is bounded by C and line segment PQ .

Use algebraic integration to find the area of R .

(5)

Worked Solution - Question 3

1. Find the line PQ

The gradient of PQ is $\frac{5-11}{4-1} = -2$, so the line through $P(1, 11)$ is $y - 11 = -2(x - 1)$.

2. Write the line equation

Thus PQ has equation $y = -2x + 13$.

3. Set up the area

From $x = 1$ to $x = 4$, the line segment is above the curve. Hence

$$R = \int_1^4 \left[(-2x + 13) - \left(3x + \frac{16}{x^2} - 8 \right) \right] dx.$$

4. Simplify the integrand

$$R = \int_1^4 (21 - 5x - 16x^{-2}) dx.$$

5. Integrate

$$R = \left[21x - \frac{5}{2}x^2 + \frac{16}{x} \right]_1^4.$$

6. Evaluate

$$R = (84 - 40 + 4) - \left(21 - \frac{5}{2} + 16 \right) = 48 - \frac{69}{2} = \frac{27}{2}.$$

Final answer

$$\frac{27}{2}.$$

Question 4

Polynomials

4: The function $f(x)$ is defined by

$$f(x) = ax^3 + bx^2 + 5x - 3$$

where a and b are constants.

Given that $(x + 3)$ is a factor of $f(x)$,

(a) show that

$$b = 3a + 2 \quad (2)$$

Given further that when $f(x)$ is divided by $(2x - 1)$ the remainder is $\frac{7}{4}$

(b) find the value of a and the value of b . (4)

(c) Using algebra, find the quotient and the remainder when $f(x)$ is divided by $(x - 2)$ (3)

Worked Solution - Question 4

1. Use the factor theorem

Since $x + 3$ is a factor, $f(-3) = 0$.

2. Show the relation

$f(-3) = -27a + 9b - 15 - 3 = 0$, so $9b = 27a + 18$ and therefore $b = 3a + 2$.

3. Use the remainder theorem

When dividing by $2x - 1$, substitute $x = \frac{1}{2}$. The remainder is $\frac{7}{4}$, so

$$f\left(\frac{1}{2}\right) = \frac{7}{4}.$$

4. Form the second equation

$\frac{a}{8} + \frac{b}{4} + \frac{5}{2} - 3 = \frac{7}{4}$. Multiplying by 8 gives $a + 2b - 4 = 14$, so $a + 2b = 18$.

5. Solve for a and b

Substitute $b = 3a + 2$ into $a + 2b = 18$: $a + 6a + 4 = 18$, so $a = 2$ and $b = 8$.

6. Write f of x

$$f(x) = 2x^3 + 8x^2 + 5x - 3.$$

7. Divide by x minus two

$$2x^3 + 8x^2 + 5x - 3 = (x - 2)(2x^2 + 12x + 29) + 55.$$

8. State the result

The quotient is $2x^2 + 12x + 29$ and the remainder is 55.

Final answer

(a) Proven.

(b) $a = 2, b = 8$.

(c) Quotient $2x^2 + 12x + 29$, remainder 55.

Question 5

Integration

x	-2	-0.5	1	2.5	4	5.5	7
y	12	4.243	1.5	0.530	0.188	0.066	0.023

The table above shows corresponding values of x and y for

$$y = 3\left(\frac{1}{2}\right)^x$$

The values of y are given to 3 decimal places as appropriate.

- (a) Using the trapezium rule with all the values of y in the given table, obtain an estimate for

$$\int_{-2}^7 3\left(\frac{1}{2}\right)^x dx$$

giving the answer to one decimal place.

(3)

Using the answer to part (a) and making your method clear, estimate

(b) (i) $\int_{-2}^7 3\left(\frac{1}{2}\right)^{x+2} dx$

(ii) $\int_{-2}^7 (2^{-x} + 2x) dx$

(3)

Worked Solution - Question 5

1. Find the strip width

The x values increase by 1.5, so $h = 1.5$.

2. Apply the trapezium rule

$$\int_{-2}^7 3\left(\frac{1}{2}\right)^x dx \approx \frac{1.5}{2} \{12 + 0.023 + 2(4.243 + 1.5 + 0.530 + 0.188 + 0.066)\}$$

3. Evaluate

This gives 18.80775, which is 18.8 to one decimal place.

4. Use the power shift

$$3\left(\frac{1}{2}\right)^{x+2} = \frac{1}{4} \cdot 3\left(\frac{1}{2}\right)^x.$$

5. Estimate part b i

$$\text{So } \int_{-2}^7 3\left(\frac{1}{2}\right)^{x+2} dx \approx \frac{1}{4}(18.80775) = 4.7019375, \text{ which is } 4.7.$$

6. Rewrite part b ii

$$\text{Since } 3\left(\frac{1}{2}\right)^x = 3 \cdot 2^{-x}, \text{ we have } 2^{-x} = \frac{1}{3} \cdot 3\left(\frac{1}{2}\right)^x.$$

7. Estimate part b ii

$$\int_{-2}^7 (2^{-x} + 2x) dx \approx \frac{18.80775}{3} + [x^2]_{-2}^7 = 6.26925 + 45 = 51.26925, \text{ which is } 51.3.$$

Final answer

(a) 18.8.

(b)(i) 4.7.

(b)(ii) 51.3.

Question 6

Laws of Logarithms

6:

In this question you must show all stages of your working.
Solutions relying on calculator technology are not acceptable.

(i) Solve

$$2\log_2(4-x) = 3 + \log_2\left(\frac{x+11}{2}\right) \quad (5)$$

(ii) The curves C_1 and C_2 with equations

$$y = 3^{2x+1} \quad \text{and} \quad y = 6 \times 3^x$$

meet at the point P .

Find the exact coordinates of P , writing your answer in the form $(\log_3 a, b)$
where a and b are integers.

(5)

Worked Solution - Question 6

1. State the domain

The logarithms require $4 - x > 0$ and $\frac{x + 11}{2} > 0$, so $-11 < x < 4$.

2. Use log laws

$$2\log_2(4 - x) = \log_2((4 - x)^2) \text{ and}$$
$$3 + \log_2\left(\frac{x + 11}{2}\right) = \log_2\left(8 \cdot \frac{x + 11}{2}\right).$$

3. Equate the arguments

$$\text{Thus } (4 - x)^2 = 4(x + 11).$$

4. Solve the quadratic

$$x^2 - 8x + 16 = 4x + 44, \text{ so } x^2 - 12x - 28 = 0. \text{ Hence } (x - 14)(x + 2) = 0.$$

5. Check the domain

The root 14 is outside the domain, so $x = -2$.

6. Equate the curves

$$\text{At } P, 3^{2x+1} = 6 \times 3^x.$$

7. Let u equal 3 to the x

$$\text{Let } u = 3^x. \text{ Then } 3^{2x+1} = 3u^2, \text{ so } 3u^2 = 6u.$$

8. Find u

Since $u > 0$, divide by $3u$ to get $u = 2$. Hence $3^x = 2$ and $x = \log_3 2$.

9. Find y

$$y = 6 \times 3^x = 6 \times 2 = 12, \text{ so } P = (\log_3 2, 12).$$

Final answer

(i) $x = -2$.

(ii) $P = (\log_3 2, 12)$.

Question 7

Sequences & Series

7:

In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.

- (i) A **geometric** series begins

$$10 + 8 + 6.4 + \dots$$

- (a) Find the sum to infinity of this series.

(2)

Given that the k th term of this series is less than 0.0005

- (b) use algebra to find the smallest possible value of k .

(3)

- (ii) An **arithmetic** series begins

$$850 + 843 + 836 + \dots$$

Given that the sum of the first n terms of this series is S_n
find the greatest possible value of S_n

(4)

Worked Solution - Question 7

Topic group

1. Identify the geometric sequence

The first term is 10 and the common ratio is $\frac{8}{10} = 0.8$.

2. Find the sum to infinity

$$S_{\infty} = \frac{10}{1 - 0.8} = 50.$$

3. Write the k th term

The k th term is $10(0.8)^{k-1}$.

4. Use the inequality

$$10(0.8)^{k-1} < 0.0005, \text{ so } (0.8)^{k-1} < 0.00005.$$

5. Solve with logarithms

Taking logarithms gives $(k - 1) \log(0.8) < \log(0.00005)$. Since $\log(0.8) < 0$,
 $k - 1 > \frac{\log(0.00005)}{\log(0.8)} = 44.381 \dots$

6. Find the smallest integer

Thus $k > 45.381 \dots$, so the smallest possible value is $k = 46$.

7. Write the arithmetic sum

For $850 + 843 + 836 + \dots$, the first term is 850 and the common difference is -7 .

8. Form S_n

$$S_n = \frac{n}{2} \{2(850) + (n - 1)(-7)\} = \frac{n}{2} (1707 - 7n).$$

9. Find the best integer n

This quadratic has its maximum near $n = \frac{1707}{14} = 121.928\dots$, so check $n = 122$.

10. Find the greatest sum

$$S_{122} = \frac{122}{2}(1707 - 854) = 61(853) = 52033.$$

Final answer

(i)(a) 50.

(i)(b) 46.

(ii) 52033.

Question 8

Trigonometric Equations

8:

In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.

- (i) Solve, for $0 \leq x < \pi$, the equation

$$3 \tan \left(2x + \frac{\pi}{5} \right) = \sqrt{3}$$

giving the answers in radians in the form $k\pi$, where k is a rational constant to be found.

(3)

- (ii) Solve, for $0 \leq \theta < 360^\circ$, the equation

$$5 \sin \theta \tan \theta = \cos \theta + 4$$

giving your answers, in degrees, to one decimal place.

(5)

Worked Solution - Question 8

1. Solve the tangent equation

$$3 \tan \left(2x + \frac{\pi}{5} \right) = \sqrt{3} \text{ gives } \tan \left(2x + \frac{\pi}{5} \right) = \frac{1}{\sqrt{3}}.$$

2. Write the general angle

$$\text{Therefore } 2x + \frac{\pi}{5} = \frac{\pi}{6} + n\pi.$$

3. Find x

$$2x = -\frac{\pi}{30} + n\pi, \text{ so } x = -\frac{\pi}{60} + \frac{n\pi}{2}.$$

4. Use the interval

$$\text{For } 0 \leq x < \pi, \text{ the possible values are } x = \frac{29\pi}{60} \text{ and } x = \frac{59\pi}{60}.$$

5. Rewrite the second equation

$$\text{Using } \tan \theta = \frac{\sin \theta}{\cos \theta}, 5 \sin \theta \tan \theta = \cos \theta + 4 \text{ becomes } \frac{5 \sin^2 \theta}{\cos \theta} = \cos \theta + 4.$$

6. Form an equation in cos theta

Multiplying by $\cos \theta$ gives $5 \sin^2 \theta = \cos^2 \theta + 4 \cos \theta$. Using $\sin^2 \theta = 1 - \cos^2 \theta$ gives $6 \cos^2 \theta + 4 \cos \theta - 5 = 0$.

7. Solve for cos theta

$$\cos \theta = \frac{-2 \pm \sqrt{34}}{6}. \text{ The negative option is less than } -1, \text{ so reject it.}$$

8. Find theta

$$\cos \theta = \frac{-2 + \sqrt{34}}{6} \text{ gives } \theta = 50.320 \dots^\circ \text{ or } 309.679 \dots^\circ.$$

9. Round the answers

To one decimal place, $\theta = 50.3^\circ$ or 309.7° .

Final answer

$$(i) x = \frac{29\pi}{60}, \frac{59\pi}{60}.$$

$$(ii) \theta = 50.3^\circ, 309.7^\circ.$$

Question 9

Applications of Differentiation

9:

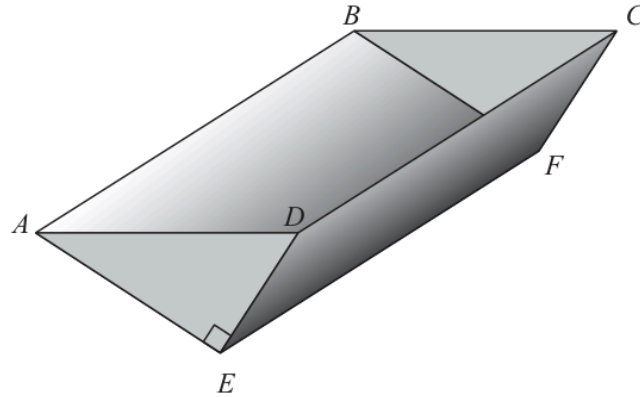


Figure 2

Figure 2 shows a sketch of an open container.

The sides $CDEF$ and $ABFE$ are rectangles.

The ends ADE and BCF are congruent (identical) right-angled triangles.

The container is made from metal of negligible thickness.

Given that

- $AE = BF = 3x$ metres
- $DE = CF = 2x$ metres
- $AB = DC = EF = L$ metres

and the capacity of the container is 12 m^3

(a) show that the area of metal used to make the container, $S \text{ m}^2$, is given by

$$S = Px^2 + \frac{Q}{x}$$

where P and Q are positive integers to be found.

(4)

Given that x can vary,

(b) use algebraic calculus to find the minimum value of S , giving your answer to one decimal place.

(5)

(c) Justify that the value of S found in part (b) is a minimum.

(2)

Worked Solution - Question 9

Topic group

1. Find the triangular end area

Each end is a right-angled triangle with perpendicular sides $3x$ and $2x$, so its area is $\frac{1}{2}(3x)(2x) = 3x^2$.

2. Use the capacity

The volume is cross-section area times length, so $3x^2L = 12$ and $L = \frac{4}{x^2}$.

3. Write the metal area

There are two triangular ends with total area $6x^2$, and two rectangular sides with total area $3xL + 2xL = 5xL$.

4. Substitute for L

$S = 6x^2 + 5x \left(\frac{4}{x^2} \right) = 6x^2 + \frac{20}{x}$. Hence $P = 6$ and $Q = 20$.

5. Differentiate

$$\frac{dS}{dx} = 12x - 20x^{-2} = 12x - \frac{20}{x^2}.$$

6. Find the stationary value of x

Set $\frac{dS}{dx} = 0$: $12x = \frac{20}{x^2}$, so $x^3 = \frac{5}{3}$.

7. Evaluate S

At $x = \sqrt[3]{\frac{5}{3}}$, $S = 6x^2 + \frac{20}{x} = 25.3029\dots$, so the minimum value is **25.3** to one decimal place.

8. Differentiate again

$$\frac{d^2 S}{dx^2} = 12 + 40x^{-3} = 12 + \frac{40}{x^3}.$$

9. Justify the minimum

Since $x > 0$, $12 + \frac{40}{x^3} > 0$, so the stationary value found is a minimum.

Final answer

(a) $S = 6x^2 + \frac{20}{x}$, so $P = 6$ and $Q = 20$.

(b) 25.3.

(c) Minimum justified.

Question 10

Proof

10:

In this question you must show detailed reasoning.

Use algebra to prove by exhaustion that,

for all positive integers m that are **not** multiples of 3, the value of

$$m^2 + 3m + 2$$

is always a multiple of 3

(4)

Worked Solution - Question 10

Topic group

1. List the possible cases

A positive integer that is not a multiple of 3 must have the form $m = 3q + 1$ or $m = 3q + 2$.

2. Case one

If $m = 3q + 1$, then

$$m^2 + 3m + 2 = (3q + 1)^2 + 3(3q + 1) + 2 = 9q^2 + 15q + 6 = 3(3q^2 + 5q + 2)$$

.

3. Case two

If $m = 3q + 2$, then

$$m^2 + 3m + 2 = (3q + 2)^2 + 3(3q + 2) + 2 = 9q^2 + 21q + 12 = 3(3q^2 + 7q + 4)$$

.

4. Conclude by exhaustion

Both possible cases give a factor of 3, so $m^2 + 3m + 2$ is always a multiple of 3.

Final answer

Proven.